POLITECNICO
MILANO 1863

Design Document

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November 23, 2024

Students&Companies - Design Document
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Contents

1 Introduction

1.1 Purpose

This document aim to provide technical information of S&C platform. This document aims to assist the development team in implementing the desired system. The document describes the system architecture and its components in detail, including interactions between them. The Design Document (DD) defines implementation, integration, and testing strategies based on priority, effort, and stakeholder impact. In general, the main different features of this document are:

- The high-level architecture
- Main components of the system
- Interfaces provided by the components
- Design patterns adopted
- Implementation, integration, and testing

1.2 Scope

Students&Companies (S&C) is a new platform that helps students to find internships, by browsing through the platform, in which they search for an internship that a company has posted, and sending their resume and initiate their application. Companies post internships allowing students to apply for those internships. Furthermore, universities monitor the application/internship of their students, and allowing students to report a complaint regarding any stage of the procedure.

1.3 Definition, Acronyms, Abbreviations

Table 1: Abbreviations

Abbreviations	**Description**

1.4 Revision History

1.5 Reference Documents

- Specification Document: "Assignment RDD AY 2024-2025.pdf"
- Course Slides

1.6 Document Structure

Section 1 Brief description of DD and introduction of purpose and scope. It also includes definitions, acronyms, and abbreviations.

Section 2 The main part of DD is this section that contains architectural design choices, including all the components, and the interfaces used for the development of the application. Also, it contains a deployment view and a runtime view. In the end, is explained the architectural patterns chosen with the other design decisions.

Section 3 Contains how should be the user interface on mobile applications.

Section 4 Contains the traceability matrix that shows which components satisfy certain requirements.

Section 5 The implementation plan, integration plan, and testing plan show how these plans are executed.

Section 6 Shows how much time is spent by each member of the group.

Section 7 Includes the reference documents.

2 Architectural Design

2.1 Overview

The architecture of this platform is structured by 3 logic layers:

1. Presentation Layer: The presentation Layer provides means of Input, allowing the users to manipulate the system Output, allowing the system to produce the results of user manipulation SAP is having Graphical User interface (SAP GUI). The SAP GUI is installed on Individual machines which act as a presentation layer.
2. Application Layer: In this layer business logic is executed. The application layer can be installed on one machine, or it can be distributed among more than one system.
3. Database Layer: The database layer holds the data. SAP supports any relational database. SAP does not provide any database. But it supports any RDBMS. The database layer must be installed on one machine or system. Major databases which are being used in SAP implementations are Oracle and DB2.

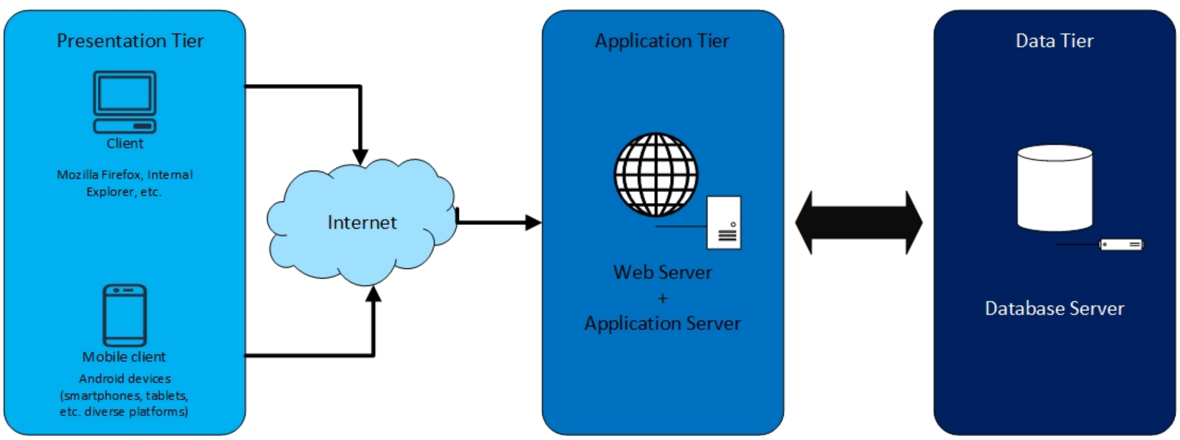

Figure 1: Three tier architecture

The following figure represents our system architecture, its high-level components, and their interaction. The client could use this by a web browser on their smartphone or computer. Then, user interactions will be processed by the application server. As well as an application server connected to the database. Finally, the application server is allowed to use APIs for Google Meets, Google Calendar.

These figures do not include all details about the architecture of the application, and in the next section, more details are represented.

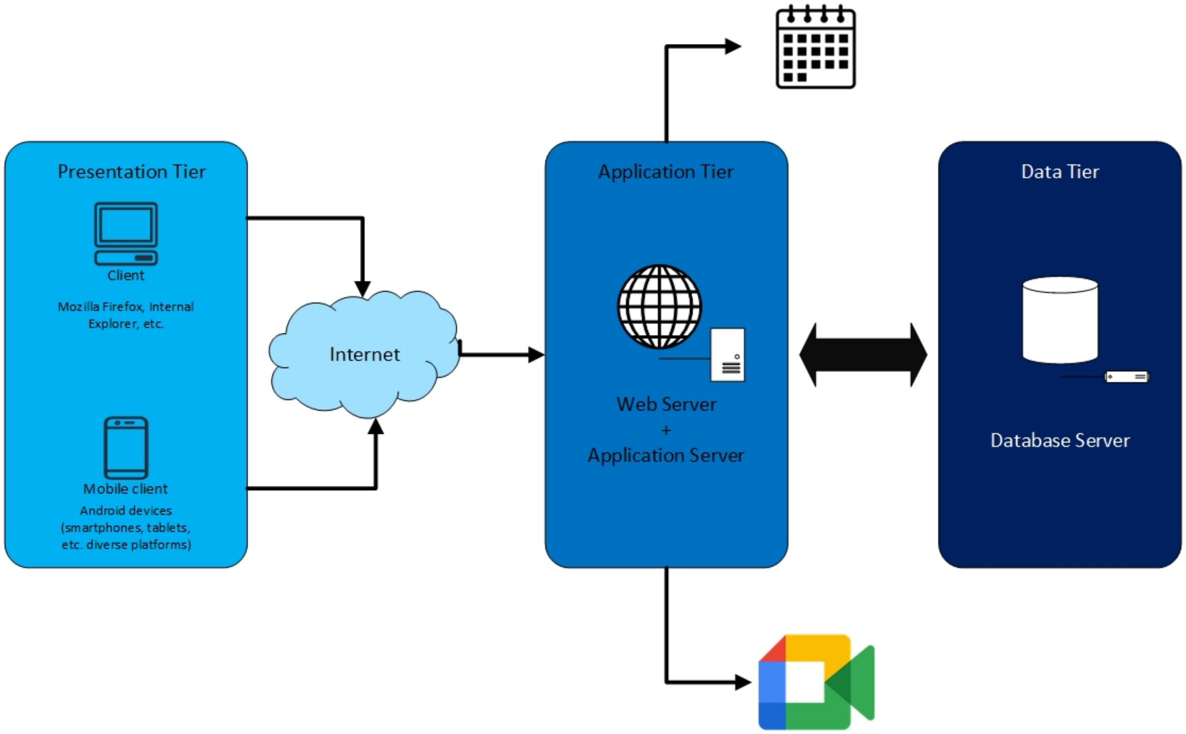

Figure 2: System Architecture

2.2 Component View

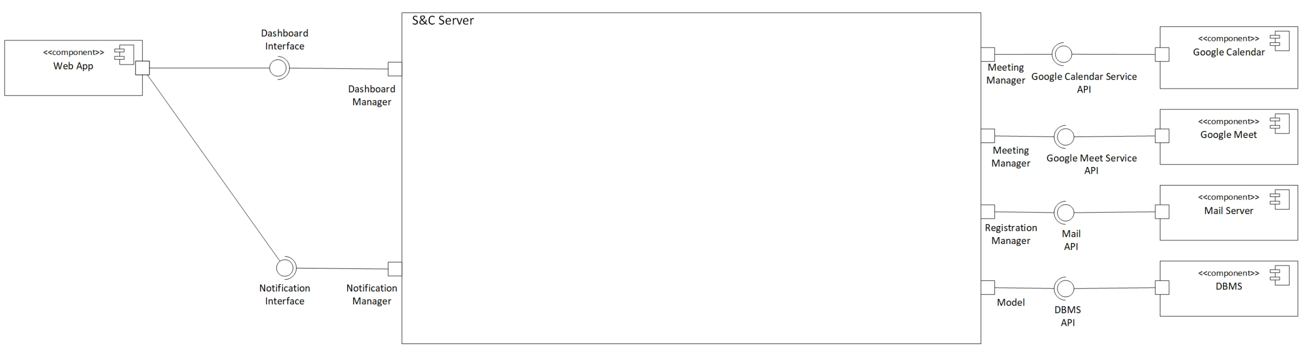

Figure 3: High Level Component Diagram

The high level component diagram above depicts S&C's external components and their communication with the server.

Web Server: The Dashboard Interface is the only way for users to communicate with the S&C Server. The S&C Server may send notifications to users via the Notification Interface, including meeting request, internship offer and etc.

DBMS: The DBMS stores all User, applications, complaints, and internships. The Model component connects to the S&C Server via the DBMS API.

Mail Server: In charge of sending emails confirming registration, the Mail Server uses the Mail API interface to connect to the S&C Server. The Registration Manager component, which manages the user registration procedure, is connected to this interface.

Google Meet: The Google Meet service is used for holding meetings in between students and company's representative for internship positions. The Google Meet service uses the Google Meet Service API interface to connect to the S&C Server. The Meeting Manager component, which manages the meetings between users, is connect to this interface.

Google Calendar: The Google Calendar service is used for keeping information about the availability regarding the meetings and internships. The Google Calendar service uses the Google Calendar API interface to connect to the S&C server. The Meeting Manager component, which manages the meetings between users, is connect to this interface.

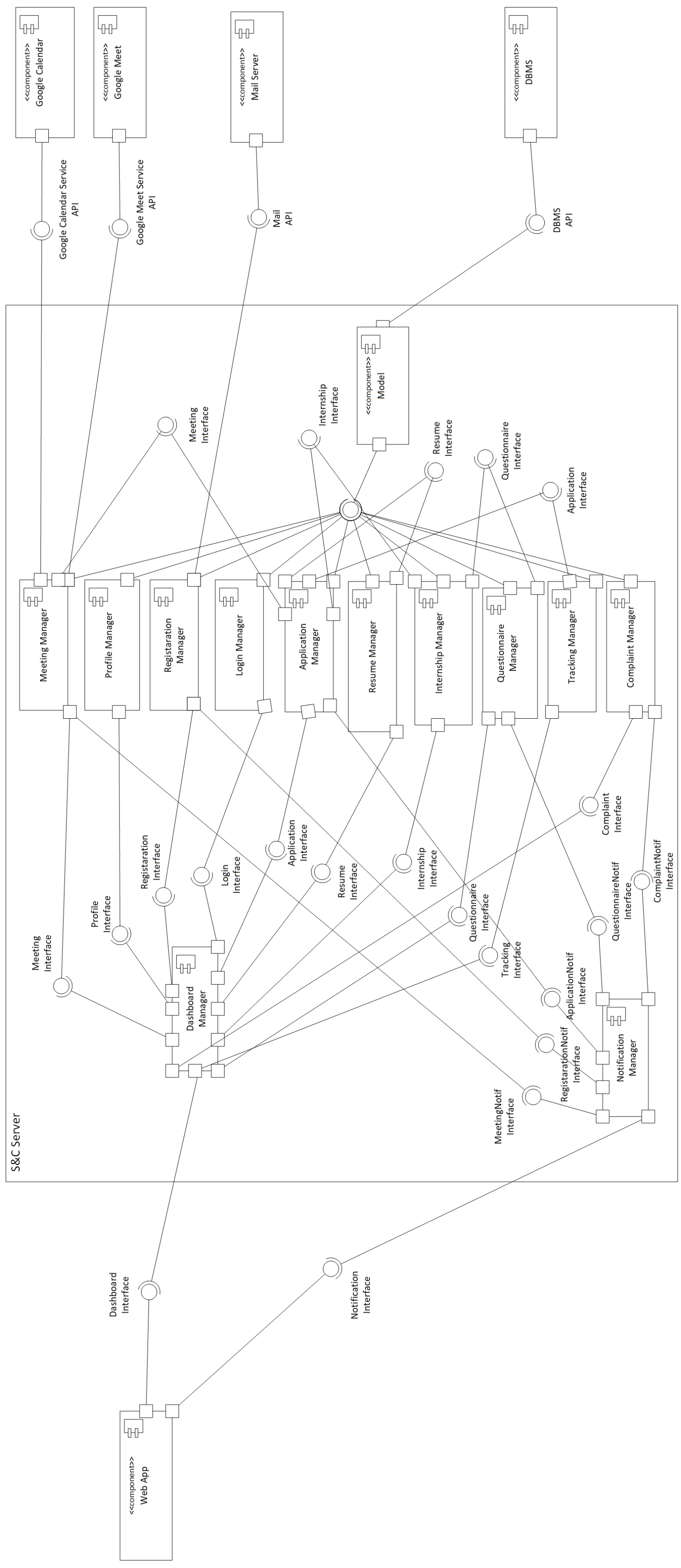

Figure 4: Low Level Component Diagram

2.3 Deployment View

2.4 Runtime View

2.5 Component Interfaces

2.6 Selected Architectural Styles and Patterns

2.7 Other Design Decisions

2.7.1 Availability

2.7.2 Scalability

2.7.3 Notification Handling

2.7.4 Ease of Deployment

2.7.5 Data Storage

3 User Interface Design

3.1 Overview

3.2 Header

3.3 User Interfaces

4 Requirements Traceability

5 Implementation, Integration and Test Plan

6 Effort Spent

References

List of Tables

List of Figures